PANJAB UNIVERSITY CHANDIGARH- 160014 (INDIA)

(Estted. under the Panjab University Act VII of 1947-enacted by the Govt. of India)

FACULTY OF SCIENCE

SYLLABI

FOR

MASTER OF COMPUTER APPLICATIONS (MCA)
(SEMESTER SYSTEM)

EXAMINATIONS 2023 – 2024

i.e.
1st Semester, November/December, 2023
2nd Semester, May/June, 2024
3rd Semester, November/December, 2023
4th Semester, May/June, 2024

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APPLICABILITY OF REGULATIONS FOR THE TIME BEING IN FORCE

Notwithstanding the integrated nature of a course spread over more than one academic year, the regulations in force at the time a student joins a course shall hold good only for the examinations held during or at the end of the academic year. Nothing in these regulations shall be deemed to debar the University from amending the regulations subsequently and the amended regulations, if any, shall apply to all students whether old or new.

PANJAB UNIVERSITY, CHANDIGARH

Outlines of Tests, Syllabi and Courses of Reading for Master of Computer Applications (MCA) (Two Year Degree Programme) for Session 2023 – 2024.

Paper Code	Paper Name	Theory & Practical Lectures	Univ. Exam. Marks	Int. Exam. Marks	Credits
FIRST YEAR					
FIRST SEMESTER					
Theory Subjects					
CS - 2110	Data and File Structures using C	4+0	80	20	4
CS - 2111	Computer Organization and Architecture	4+0	80	20	4
CS - 2126	Mathematical Structures and Linear Programming	4+0	80	20	4
CS - 2113	Relational Data Base Management Systems	4+0	80	20	4
CS – 2114	Operating Systems	4+0	80	20	4
Practical Subjects					
PR – 2110	Practical Based on **CS - 2110** Programming in C & Data Structures (Minor Project)	0+4	80	20	4
PR – 2113	Practical Based on **CS – 2113 and CS- 2114** Linux and RDBMS (SQL Server/Oracle/My SQL) (Minor Project)	0+4	80	20	4
SECOND SEMESTER					
Theory Subjects					
CS – 2115	Object Oriented Programming (Through C++ and Java)	4+0	80	20	4
CS – 2116	Computer Networks and Security	4+0	80	20	4
CS – 2124	Artificial Intelligence and Soft Computing	4+0	80	20	4
CS – 2118	Web Technologies and Python Programming	4+0	80	20	4
CS – 2119	Analysis and Design of Algorithms	4+0	80	20	4
Practical Subjects					
PR – 2115	Practical Based on **CS – 2115** Object Oriented Programming (C++ and Java)	0+4	80	20	4
PR – 2118	Practical Based on **CS - 2118 & CS - 2124** Web Technologies and Python Programming & Artificial Intelligence and Soft Computing	0+4	80	20	4

Note : Each student is required to undergo 04 to 06 weeks Summer Training at the end of Second Semester. The Internal Assessment Marks for each practical will be based on Minor Project.

SECOND YEAR					
THIRD SEMESTER					
Theory Subjects					
CS – 2120	Interactive Computer Graphics	4+0	80	20	4
CS – 2121	Theory of Computations	4+0	80	20	4
CS – 2122	Advance JAVA and Network Programming	4+0	80	20	4
CS – 2123	Mobile Communication and Application Development	4+0	80	20	4
CS – 2117	Software Engineering and Project Management	4+0	80	20	4
Practical Subjects					
PR – 2120	Practical Based on **CS – 2120 and CS-2123** Interactive Computer Graphics and Mobile Communication and Application Development	0+4	80	20	4
PR – 2122	Practical Based on **CS - 2122** Advance JAVA and Network Programming	0+4	80	20	4
FOURTH SEMESTER					
CS – 2125	**PROJECT WORK**				
	The Project period will be of 16 to 20 weeks duration. The Project will involve development of application/system software in industries, commercial or scientific environment. It will carry 400 marks.		320	80	16

SYLLABUS AND COURSE OF READINGS

FIRST SEMESTER

Paper Title: Data and File Structures using C

Paper Code: CS - 2110 **Max. Marks : 80 Time : 3 Hrs.**
Course Duration: 60 Lectures of one hour each.

Objective: The objective of this course is to assist students in developing the logic for solving a given problem using the procedure oriented language.

Note :
(i) The Question paper will consist of four units.
(ii) Examiner will set total of **nine** questions comprising two questions from each unit and one compulsory question of short answer type covering whole syllabi and having equal distribution of marks from all the units.
(iii) The students are required to attempt one question from each unit and the compulsory question.
(iv) All questions carry equal marks.

Course Outcome
After successful completion of course the students will be able to:

- Understand the fundamental concepts of C programming language
- Familiarize with common data structures (such as arrays, linked lists, stacks, queues, graphs, trees, files etc.).
- Select, apply and implement the appropriate data type and data structure for a given problem using C.
- Learn further through investigation of program behaviour and debugging efforts.

UNIT I

Review of C language concepts: Structure of a C program, Variables, Constants, Keywords, Data types, Operators, Evaluation of expressions, typecasting, Input/output statements, Assignment statements, Control statements: if-else, switch, while, do-while, for, nested loops, break, continue, goto statements.

Functions: Declaration, Definition, Function prototype, Passing arguments, Recursion, Use of library functions, Storage classes: automatic, external and static variables.

Arrays: Defining and processing arrays, Passing an array to a function, Using single and multi dimensional arrays.

UNIT II

Pointers and Strings: Pointer declaration, Operations on pointers, Passing pointers to functions, Array of pointers, pointer to array; Dynamic memory allocation; String declaration, Operations on strings.

Structure and Union: Defining and processing structure, Passing structure to a function, Arrays of structures, Pointers and structures.

Files Handling: Concept of files, file opening modes, opening and closing of a file, reading from a file, writing onto a file.

UNIT III

Review of Data Structure: Implementing array operations using C (insertion, deletion, traversing, reversing, searching, sorting, merging two arrays and matrix operations); Implementing sequential and linked representation of stacks and queues using C and performing various operations. Implementing various searching (Linear, Binary) and sorting algorithms (Bubble, Insertion, Selection, Quick, Merge) in C.

Linked Lists: Implementing singly linked list, circular linked list and doubly linked list using C and performing various operations.

UNIT IV

Trees: Terminology, Binary tree, Complete binary tree, Binary search tree, Traversal Techniques: in-order, pre-order, post -order; implementing operations in C (insertion, traversal, searching, deletion).

Graphs: Terminology, Operations: Insertion, deletion, traversal: DFS, BFS; Minimum cost spanning trees: Kruskal's algorithm; implementation in C.

File structures: Sequential Access, Direct Access, Indirect Sequential Access, Hashing techniques for direct files, implementation in C.

Suggested Readings:

1. Kanetkar ,Yashavant, 2021, Let Us C, BPB Publications.
2. Cooper, Mullish, 1998: The Spirit of C, An Introduction to Modern Programming, JaicoPubl. House, New Delhi.
3. Kerninghan, B.W. &Ritchie, D.M.: The C Programming Language, Prentice Hall of International, 2015.
4. Gottfried, B.: Theory and Problems of Programming in C, Schaum Series,1996.
5. Horwitz, E., and Sahni, S., 2008: Fundamentals of data structures in C, Computer Science Press.
6. Aho, A. V., Hopcroft, and Ullman, J.E., 2002: Data structures and algorithms, Addison Wesley.
7. Tanenbaum, A. M. and Augenstein, M.J., 2019: Data structures using C, Prentice Hall
8. Lipschutz, Seymour, 2014: Theory & problems of data structures, Schaum Series.
9. Deshpanday: C and data structures, Wiley India Pvt. Ltd,2003.

Paper Title: Computer Organization and Architecture

Paper Code : CS - 2111 **Max. Marks : 80 Time : 3 Hrs.**
Course Duration : 60 Lectures of one hour each.

Objectives: Examine the operation of the major building blocks of a computer system on the concepts that are the basis of the current computers such as microprocessors. To introduce Parallel & Multiprocessing Environment, Superscalar and Super pipeline Design.

Note :
(i) The Question paper will consist of four units.
(ii) Examiner will set total of **nine** questions comprising two questions from each unit and one compulsory question of short answer type covering whole syllabi and having equal distribution of marks from all the units.
(iii) The students are required to attempt one question from each unit and the compulsory question.
(iv) All questions carry equal marks.

Course outcomes:
After successful completion of course the students will be able to:

- Solve the functions and expression and built various circuits using logic gates.
- Demonstrate computer architecture concepts related to design of modern processors, memories and I/Os.
- Understand the theory and architecture of central processing unit and functionality of central processing unit. I/O and memory organization.
- Understand the concepts of parallel processing, pipelining and interprocessor communication.

UNIT I

1. **Basic Building Blocks:** Digital logic gates, Boolean algebra: functions and expressions, Truth tables, simplification of Boolean functions, K-maps for 3 and 4 variables.

2. **Combinational and Sequential Logic Design:** Half-adder, Full adder, Encoder, Decoder, Multiplexer. Sequential circuits – Flip-flops (D, RS, JK, T, and Master-Slave); Registers: Buffer, Shift and Controlled shift registers; Counters: Binary, Ripple, Ring Counter.

UNIT II

3. **Register Transfer and Micro-operations:** Register Transfer Language, Bus and Memory Transfer, Logic and Shift micro-operations.

4. **Computer Organization:** Microprocessor Organization; Instruction codes, Instruction cycle, Instruction formats, Processing Unit Design: one, two and three- bus Organization, Control Unit: Hardwired Control and Micro-programmed control approach.

UNIT III

5. **Overview of Memory Subsystem Organization and Interfacing :** Types of Memories:- RAM, ROM, Cache and Associative memories. I/O Subsystem Organization: I/O interface, DMA, Priority interrupts.

6. **Micro Processor Architecture:** Microcomputer Architecture, Structure of 8086/8088 CPU, The Bus Interface Unit, Execution Unit (EU); Registers, Addressing modes, Programmers model of a machine, Machine vs. Assembly Language, Assembler : One Pass and Two Pass Assembler

UNIT IV

6. **Parallel and Multiprocessing Environment:** Introduction to parallel processing, RISC Architecture, Vector and Array processing, Multiprocessing concepts, Multiprocessor organization, Memory and Resource sharing, Inter-processor communication and Synchronization.

7. **Superscalar and Super Pipeline Design:** Introduction to pipelining, Superscalar Pipeline Design, SIMD(Single Instruction and Multiple Data Stream) Array Processors, SIMD Computer Architecture, MIMD(Multiple Instruction and Multiple Data Stream) Architecture, Difference between SIMD and MIMD.

Suggested Readings:

1. Mano, M.M.: Computer System Architecture, Prentice Hall of India
2. V. Rajaraman, T. Radhakrishnan: Computer Organization and Architecture, PHI Learning Pvt. Ltd
3. Faye A. Briggs: Computer Architecture and Parallel Processing, McGraw-Hill International Editions
4. William Stallings: Computer Organization and Architecture: Designing for Performance, Pearson
5. John d. Carpinelli: Computer Systems Organization & Architecture, Addison Wesley.
6. Hayes : Computer Architecture and Organization, McGraw-Hill International Edition.
7. SchaumSeries : Introduction to Microprocessors .

Paper Title: Mathematical Structures and Linear Programming

Paper Code: CS – 2126 **Max. Marks: 80 Time: 3 Hrs.**
Course Duration: 60 Lectures of one hour each.

Objective: To provide basic knowledge about mathematical structures required for various computer science courses.

Note :
(i) The Question paper will consist of four units.
(ii) Examiner will set total of **nine** questions comprising two questions from each unit and one compulsory question of short answer type covering whole syllabi and having equal distribution of marks from all the units.
(iii) The students are required to attempt one question from each unit and the compulsory question.
(iv) All questions carry equal marks.

Course Outcomes:
After successful completion of course the students will be able to:

- Develop the ability to think, read and write Algorithms effectively. The functions understanding including comparison of functions will help in programming concepts.
- Understand important logical concepts as tautologies, contradictions, Laws in Logic, logical equivalence, formulating quantified statements, truth tables and truthfulness of a compound statement.
- Understand numerical, mathematics contents and solutions to Scientific Problem like solution to non-linear equations using Bisection method, False position method, Newton-Raphson method, Secant method.
- Understand various methods of interpolation, numerical integration and Linear and Non-Linear Regression which will help the students to understand important concepts and methods in numerical analysis.

UNIT I

1. **Overview:** Counting, Pigeon-hole principle, generating functions, Mathematical induction, summation formulas and its properties, Asymptotic notations, monotonicity.

2. **Functions:** Comparison of standard functions as floors and ceilings, Polynomials, exponentials, Logarithms, Composite functions and their range, Domain, Functions for computer science like characteristic function, Hashing function.

UNIT II

3. **Mathematical Logic:** Introduction, Conjunction, Disjunction and negation, Propositions and truth table, Tautologies and contradictions, Equivalence of formulas, Duality law, Normal forms: Disjunctive Normal form, Conjunctive Normal form, Predicate Calculus: Predicates, the statement function, variables and quantifiers, predicate formulas. Methods of proof (Inference Theory).

UNIT III

4 **Probability :** Discrete probability. Expected values and variance.
Relations : Properties, Combining relations, Closures, Equivalence, partial ordering.
Graphs : Directed, undirected graphs, walks, paths, trails, connected graphs, regular and bipartite graphs, cycles and circuits, eccentricity of a vertex. Hamiltonian and Eulerian graphs, Planar graphs.

UNIT-IV

5. Linear Programming: Mathematical Model; Introduction to LPP, Graphical Method, Simple methods to solve LPP and applications.

Suggested Readings:

1. Kolman, Busby, Ross : Discrete Mathematical Structure, PEARSON Education, 5th Edition.
2. C. L. Liu, Elements of Discrete Mathematics, Mcgraw-Hill College,
3. Rajaraman, V., Computer Oriented Numerical Methods, Prentice Hall of India.
4. Seymor Lipschutz, Marc Lipson, Discrete Mathematics, McGraw Hill Professional,
5. J. P. Tremblay & R. Manohar. , Discrete Mathematical Structures with Applications to Computer Science, McGraw-Hill, Inc.
6. K.E. Atkinson, W. Han, Elementary Numerical Analysis, 3rd Edition, Wiley

Paper Title: Relational Database Management Systems

Paper Code: CS - 2113 **Max. Marks: 80 Time: 3 Hrs.**
Course Duration: 60 Lectures of one hour each.

Objective: Objective: The course aims at providing the students insight on basic DBMS fundamentals principles and practices. Students will further learn RDBMS concepts and implement its operations for database creation and manipulation through SQL and PL/SQL.

Note :

(i) The Question paper will consist of four units.
(ii) Examiner will set total of **nine** questions comprising two questions from each unit and one compulsory question of short answer type covering whole syllabi and having equal distribution of marks from all the units.
(iii) The students are required to attempt one question from each unit and the compulsory question.
(iv) All questions carry equal marks.

Course Outcomes:
After successful completion of course the students will be able to:

- Describe the fundamental elements of Relational Database Management Systems, basic concepts of Relational data model, Entity-relationship model, Relational database design, Relational Algebra SQL.
- Design ER-models and convert the ER-model to relational tables, Normalize database and populate relational database.
- Manipulation of databases using SQL commands.
- Understanding block structure of PL/SQL, functions and procedures using PL/SQL and perform Cursor and Trigger Management.

UNIT I

Basic Concepts: DBMS, 3-tierArchitecture, Data independence, Conventional data models: Hierarchical, Network, Relational. Entity- relationship model.

Normalization: Normal forms- INF, 2NF, 3NF, BCNF, Difference between 3NF and BCNF, Multivalued dependencies and join dependencies, 4NF, 5NF, Difference between 4NF and 5NF.

UNIT II

Relational Algebra and Relational Calculus: Relational Algebra: Operations- Union, Intersection, Difference, Cartesian product, Projection, Selection, Division and Relational algebra queries.

Transaction and Concurrency Control: Concept of Transaction, ACID properties, Serializibility, States of transaction, Concurrency control : Locking techniques ,Time stamp based protocols, Granularity of data items, Crash Recovery and Backup: Atomicity, Log base recovery,

UNIT III

Structure Query Language (SQL) : Introduction to SQL, Oracle server and oracle database, Starting SQL *Plus,: DDl, DML, DCL
Querying database tables, Conditional retrieval of rows, Working with nullvalues, Matching a pattern from a table, Ordering the result of a query, Aggregate Functions, Grouping the result of a query, ROLLUP operation: Getting sub totals, CUBE operation :Getting cross tabs, Command summary of SQL *Plus editor, Querying multiple tables : EquiJoins, Cartesian Joins, Outer Joins, Self Joins; SET Operators :Union, Intersect, Minus; Functions : Arithmetic functions, Character functions, Date functions, Group functions. VIEW : Manipulating the Base table, Rules of DML Statements onJoin Views, Dropping a VIEW.

Database security and privileges, GRANT command, REVOKE, Commit, Rollback

UNIT IV

PL/SQL-I:

Introduction to PL/SQL, PL/SQL block structure, PL/SQL Architecture, Fundamentals of PL/SQL, PL/SQL Data types, Variables and Constants, Scope and visibility of a variable, Assignments and expressions, Operator precedence, Referencing Non PL/SQL variables, Built-in-functions, Conditional and iterative control, SQL within PL/SQL, Writing PL/SQL code, Composite data types.

PL/SQL-II:

Cursor management in PL/SQL, Cursor manipulation, Implicit cursor attributes, Exception handling in PL/SQL; Predefined exceptions, User defined exceptions.

Subprograms in PL/SQL, Advantages of subprograms, Procedure, Functions, Dropping a procedure, Dropping a function, Dropping a package, Using stored function in SQL statements, Database trigger, Types of triggers, Dropping triggers, Packages.

Suggested Reading:

1. Desai, B.C., 1993: An Introduction to Database Systems, Galgotia Publ. Private Ltd.
2. Date, C.J.: Data Base Systems, Vols. I & II, Narosa Publications.
3. Ivan Bayross : PL/SQL The Programming Language of ORACLE, (BPB Publication)
4. Mukhi, Vijay 1992: Mastering Oracle 6.0, BPB Publications.
5. Database system concepts :Korth
6. Principles of Database Management: James Martin
7. Computer Database organization : James Martin
8. Fundamentals of Database Systems: ElmasriNavathe

Paper Title: Operating System

Paper Code: CS - 2114 **Max. Marks : 80 Time : 3 Hrs.**
Course Duration: 60 Lectures of one hour each.

Objectives: To understand the various concepts of Operating System like process management, synchronization, deadlocks, storage and memory management.

Note :
(i) The Question paper will consist of four units.
(ii) Examiner will set total of **nine** questions comprising two questions from each unit and one compulsory question of short answer type covering whole syllabi and having equal distribution of marks from all the units.
(iii) The students are required to attempt one question from each unit and the compulsory question.
(iv) All questions carry equal marks.

Course Outcomes:
After successful completion of course the students will be able to:

- Describe the important computer system resources and the role of operating system in their management policies and algorithms.
- Understand the process management policies and scheduling of processes by CPU
- Evaluate the requirement for process synchronization and coordination, memory management handled by operating system also cater-policies with respect to different storage.
- Learn to use features of Linux for programming and administration

UNIT I

1. Introduction to Operating System:
Introduction to Operating System, Its need and services; Different types of operating systems: batch, multi-programmed, time sharing, real time, distributed, parallel.

2. Process Management and Synchronization
Process: Process state, Process control block, Inter process communication: Shared memory systems, Message passing systems; CPU Scheduling- scheduling criteria, Preemptive & non-pre-emptive scheduling, Scheduling Algorithms: FCFS, SJF, RR and Priority.

Synchronization: Critical section problem, Peterson's solution, Synchronization hardware, Semaphores: Mutual exclusion, Producer-consumers, Reader-writers problem; Dining philosophers Problem;

UNIT II

3. Deadlocks: Introduction to deadlocks, Conditions for deadlock, Resource allocation graphs, Deadlock Detection and Recovery, Deadlock Avoidance, Deadlock Prevention

4. Memory Management and Disk Scheduling:
Logical vs. Physical address space, Memory management without swapping; Swapping; Contiguous memory allocation, Paging, Segmentation, Segmentation with paging; Virtual Memory, Demand paging, Page replacement algorithms (FIFO, Optimal, LRU); Thrashing.

Disk Scheduling algorithms (FCFS, SSTF, SCAN, C-SCAN, and LOOK), Disk Management.

UNIT III

5. Features of LINUX Operating System:
Linux File System - ext2 and ext3 file system architecture, blocks and fragments, I-node directory structure; File and Directory management features; Process management features, Text manipulation features; Regular expressions, Features of Bourne Again Shell (BASH).

6. Linux Shell Scripting:
Programming in the Bourne Again Shell (BASH); Wild card characters, Simple shell scripts; Shell variables; shell programming constructs; interactive shell scripts; Advanced features.

UNIT IV

7. Linux System Administration:
Definition, Role and functions of a system administrator, Booting the system, Run Levels in Linux, Maintaining user accounts, File System and Special Files, Backups and restoration.

8. Linux System Calls:
C as System Programming Language; I/O system calls – umask(); create(); open(); read(); write(); lseek(); dup(); link(); access(); chmod(); chown(); Process management system calls – fork(); getpid(); getppid(); exit(); wait(); sleep() ; Signal system calls – kill(); signal(); Exec System Calls – execl();execlp();execv();execvp().

Suggested Readings:

1. Peterson, James, L. and Silberschatz, A., 1985: Operating System Concepts, Wiley Publ. Comp.
2. Dhamdhere,D M:Operating Systems-A concept based approach,Mc Graw Hill.
3. Deitel, H.M., 1984: An Introduction to Operating System, Addison-Wesley Publ.Comp.
4. Milenkovic, M., 1987: Operating System – Concepts and Design, McGraw Hill International Editions.
5. Richie: Operating System, BPB.
6. Hansen PerBrineh, 1978: Operating System Principles, Prentice Hall India.
7. Madnick and Donovan: Operating System, McGraw Hill Book Co.
8. Joshi, R.C.: Operating Systems, Wiley India Pvt. Ltd.
9. Parker, Tim : Linux Unleashed, Latest Edition, Techmedia.
10. Stones, Richard and Mathew Neil: Beginning Linux Programming, 3rd Edition, Wrox.

PRACTICAL EXAMINATIONS (SESSION 2022-2023)
The Practical examination will be conducted for each of the following:

Title : Programming in C & Data Structures (Minor Project)
Paper Code : PR - 2110 (Based on CS - 2110)

Time : 3 Hrs.
Max. Marks : 100
External : 80
Internal : 20

Note : i) Each student is required to undergo 04 to 06 weeks Summer Training at the end of Second Semester.
The Internal Assessment Marks for each practical will be based on Minor Project.
ii) Paper will be set at the time of examination. Due weight-age may be given to the practical note-book and Assignments in evaluation.

Title : Linux and RDBMS (SQL Server/Oracle/My SQL) (Minor Project)
Paper Code : PR - 2113 (Based on CS – 2113)

Time : 3 Hrs.
Max. Marks : 100
External : 80
Internal : 20

Note : i) Each student is required to undergo 04 to 06 weeks Summer Training at the end of Second Semester.
The Internal Assessment Marks for each practical will be based on Minor Project.
ii) Paper will be set at the time of examination. Due weight-age may be given to the practical note-book and Assignments in evaluation.

SEMESTER II

Paper Title: Object Oriented Programming (Through C++ and Java)

Paper Code : CS - 2115 **Max. Marks : 80 Time : 3 Hrs.**

Course Duration : 60 Lectures of one hour each.

Objectives : This course will expose the students to the features in C++ and Java as well as help them in designing software applications using the object oriented paradigm of C++ and Java.

Note :

(i) The Question paper will consist of four units.

(ii) Examiner will set total of **nine** questions comprising two questions from each unit and one compulsory question of short answer type covering whole syllabi and having equal distribution of marks from all the units.

(iii) The students are required to attempt one question from each unit and the compulsory question.

(iv) All questions carry equal marks.

Course Outcomes:

After successful completion of course the students will be able to:

- Describe the meaning of the Object-Oriented Paradigm, and create class hierarchies using the Object-Oriented Design process
- Design and Implement C++ programs for complex problems, making good use of the features of the language such as classes, inheritance and templates
- Design and Implement JAVA programs through various Applications / Minor Project
- Learn further through individual investigation of program behavior and debugging efforts and will develop a professional attitude towards the programming enterprise.

UNIT I

1. **Introduction to Object Oriented Programming :** Characteristics of OOPs, Dynamic binding, Message Passing, Tokens, Expressions, Data Types, Variables, Operators, Control Statements, Arrays, string handling.

UNIT II

2. **Fundamental of OOPs :** Classes, Objects, Constructors, Destructors, new and delete operators, Inheritance, Access modes, Methods Overloading, friend functions, Pass by value vs. Pass by Reference, Operator overloading, Dynamic Polymorphism: Virtual functions, Pure virtual functions, Abstract class; Exception Handling, File Handling, Templates.

UNIT III

3. **Java** : Introduction, JVM, Byte Code, Data Types, Variables, Arrays, Operators, Control Statements, Classes, Objects, Overloading Methods, Member Access and Inheritance, Method Overriding, Using super and final, Defining a Package, Understanding CLASSPATH, Importing Packages, Interface and its implementations.

UNIT IV

4. **Multithreaded Programming :** Lifecycle of thread, Creating a thread, Implement run-able, Extending Thread, Thread Priorities; Using Synchronized Methods, Suspending, Resuming and Stopping Threads; Applet Resource Space : Applet lifecycle, Applet class, Creating applets; AWT: Using AWT Controls, Layout Managers.

Suggested Readings:

1. Balaguruswamy, E.: Object Oriented Programming with C++, TMH.
2. Lafore, Robert: OOP in Turbo C++, Galgotia.
3. Strostrup: The C++ Programming Language, Addison Wesley.
4. Parsa, N.R.: OOPS with C++ from the Foundation, Wiley India Pvt. Ltd.
5. Gaddis, Tonny : Starting out with C++, 3rd Edition, Wiley India Pvt. Ltd.
6. Gaddis, Tonny: Starting out with Object Oriented Programming in C++, 3rd Edition, Wiley India Pvt. Ltd.
7. Al Steven: Al Steven's C++ Programming, 7th Edition, Wiley India Pvt. Ltd.
8. Nicolai: Object Oriented Programming in C++, Wiley India Pvt. Ltd.
9. Schildt, Herbert: The Complete Reference Java 2, TMH.
10. Balaguruswamy, E.: Programming with Java, TMH.

Paper Title: Computer Networks and Security

Paper Code : CS - 2116 **Max. Marks : 80 Time : 3 Hrs.**
Course Duration : 60 Lectures of one hour each.

Objectives : As part of this course, students will be introduced to computer networks and data communication paradigms, about network models and standards , network protocols and their use, wireless technologies.

Note :
(i) The Question paper will consist of four units.
(ii) Examiner will set total of **nine** questions comprising two questions from each unit and one compulsory question of short answer type covering whole syllabi and having equal distribution of marks from all the units.
(iii) The students are required to attempt one question from each unit and the compulsory question.
(iv) All questions carry equal marks.

Course Outcome
After successful completion of course the students will be able to:

- Understand the fundamental concepts of computer networking
- Understand the services of Physical layer, Data Link layer, Network layer, Transport Layer and Application layer
- Familiarize with significance and techniques of network security, Cryptography, and Data Encryption Standard.
- Understand Web security, Firewalls and Proxy Firewall.

UNIT I

1. **Introduction:** Review of Data Communication concepts; Wired and wireless networks; Reference Models: OSI, TCP/IP and their comparison, Services of Physical layer and Data Link layer.

2. **Network Layer:** Logical Addressing: IPv4 and IPv6; Packet Formats and their comparison; Routing algorithms: Distance Vector, Link State Routing, Hierarchical Routing, Broadcast and Multicast Routing. Congestion Control: Principles of Congestion Control, Congestion prevention policies, Congestion control algorithms (Leaky Bucket, Token Bucket, Choke Packet, Load shedding).

UNIT II

3. **Transport Layer:** Process to Process delivery, addressing, establishing and releasing connection, flow control, buffering, Transport Protocols (TCP and UDP).

4. **Application Layer:** WWW and HTTP; File Transfer Protocol; Domain Name System; SMTP.

UNIT III

5. **Network Security- I:** Introduction, Need for Security, Security Approaches, Principles of security; Types of Attacks: Passive and Active attacks; Cryptography Techniques: Introduction, Plain text and Cipher Text, Substitution Techniques, Transposition Techniques.

6. **Encryption and Decryption** – Symmetric key cryptography; Data Encryption Standard (DES), Advanced Encryption Standard (AES), Asymmetric Key cryptography; RSA Algorithm.

UNIT IV

7. **Web Security:** Secure Sockets Layer (SSL): Architecture, Protocols, SSL Message formats, Transport Layer Security (TLS); Time Stamping Protocol(TSP); HTTPS; Secure Electronic Transaction (SET).

8. **System Security:** Intruders, Intrusion detection, password management; Firewalls, Packet Filter Firewall, Proxy Firewall.

Suggested Reading:

1. Tanenbaum, Andrew S., 2009: Computer Networks (5th Edition), PHI.
2. Forouzan, B. A., 2009: Data Communications and Networking, Fourth Edition, Tata McGraw Hill.
3. Atul Kahate, Cryptography and Network Security (2nd Edition), Tata McGraw Hill.
4. William Stalling: Cryptography and Network Security, Principles and Practise, 7th Edition, Pearson.
5. Forouzan: Cryptography and Network Security, (3rd Edition), Tata McGraw Hill.

Paper Title: Artificial Intelligence and Soft Computing

Paper Code: CS – 2124 **Max. Marks: 80 Time: 3 Hrs.**

Course Duration: 60 Lectures of one hour each.

Objective: The objective of this course is to introduce the concepts of Artificial Intelligence (AI), Applications and Machine Linearning with Soft Computing Technologies.

Note :

(i) The Question paper will consist of four units.
(ii) Examiner will set total of **nine** questions comprising two questions from each unit and one compulsory question of short answer type covering whole syllabi and having equal distribution of marks from all the units.
(iii) The students are required to attempt one question from each unit and the compulsory question.
(iv) All questions carry equal marks.

Course Outcome

After successful completion of course the students will be able to:

- Understand various heuristic search techniques and able to apply to various complex problems in Artificial Intelligence.
- Able to analyze and represent knowledge using knowledge representation schemes as Predicate logic and semantic; networks, develop expert systems and natural language processing systems.
- Understand the concept of Fuzzy logic, Genetic algorithms and their applications.
- Design and develop solutions to real world problems in various fields as medicine and engineering using relevant Soft Computing Techniques.

UNIT I

1. Introduction to Artificial Intelligence (AI) and Heuristic Search Technologies :
Introduction to AI technique, Turing test, History and developments in AI, Applications of AI, State space representation, Production systems and problem characteristics.

2. Introduction to heuristic search : Generate and test, Hill Climbing, Best First search, A*, Problem reduction: AO*, Constraint satisfaction and Means-ends-analysis techniques.

UNIT II

3. Knowledge representation and resolution principle : Expert system characteristics, Architecture and examples, Introduction to National Language processing (NLP), NLP Process.

4. Introduction to Soft Computing : Neural networks, Fuzzy logic, Genetic algorithms, Hybrid systems, Applications of soft computing.

5. Artificial Neural Network: Characteristics of neural networks; Basic Building Blocks of artificial neural network; Connections; Learning methods, Activation functions; Neural network architectures.

6. Supervised Learning Networks : Introduction to supervised learning, Architecture and training algorithms for perception network and Back propagation networks.

UNIT III

7. Unsupervised Learning Networks : Introduction to unsupervised learning, Architecture and training algorithms for Kohonen self-organizing maps.

8. Classification Techniques : Introduction and Techniques : (k-nearestneighbors, Random Forest, Support Vector Machines)

9. Clustering and Data Reduction : Clustering using K-means, Data reduction using Principle Component Analysis

UNIT IV

10. Fuzzy Logic and Control Systems:
Introduction and Applications of Fuzzy logic, Fuzzy Techniques, Membership functions: Features & Methods of membership value assignment, Defuzzification methods. Architecture Logic Control (FLC) systems.

11 .Genetic Algorithms:
Basic terminologies in genetic algorithm: Genes, Fitness and populations, General Genetic Algorithm, Operations in genetic Algorithm: Selection, Crossover and Mutation, Applications of Genetic Algorithms.

Suggested Readings:

1. Rich Elaine and Knight Kevin Shiva Shankar B Nair: Artificial Intelligence, Third Edition, Tata-McGraw Hill, 2017.
2. Patterson, Dan W.: Introduction to Artificial Intelligence and Expert systems, Pearson Education, 2015
3. S.N. Sivanandam and S.N. Deepa: Principles of Soft Computing, Second Edition, John Wiley.
4. Russel & Norvig: Artificial Intelligence, Pearson.
5. S. Rajasekaran and Pai: Neural Networks, Fuzzy Logic, and Genetic Algorithms, PHI.
6. Rao and Rao: C++ Neural Networks & Fuzzy Logic, BPB.
7. S.N. Siranandam, S. Sumathi and S.N. Deepa : Introduction to Neural Networks using MATLAB, McGraw Hill, 2014.

Paper Title : Web Technologies and Python Programming
Paper Code : CS – 2118 **Max. Marks : 80 Time : 3 Hrs.**
Course Duration : 60 Lectures of one hour each.

Objectives : To create familiarity with various Web Technologies which can be used at Client side / Sever side and encourage web applications development using them.

Note :
(i) The Question paper will consist of four units.
(ii) Examiner will set total of **nine** questions comprising two questions from each unit and one compulsory question of short answer type covering whole syllabi and having equal distribution of marks from all the units.
(iii) The students are required to attempt one question from each unit and the compulsory question.
(iv) All questions carry equal marks.

Course Outcome
After successful completion of course the students will be able to:

- Understand the basics of web development and the role of HTML, CSS, and JavaScript in creating web pages.
- Understand the usage XML and AJAX to create web applications that can retrieve data from a server without requiring a page refresh.
- Use Angular JS to create reusable components and manage the flow of data in a web application.
- Use Python programming language to create web applications using frameworks such as Django.

UNIT I

1. Introduction: World Wide Web, Web Browser, Web Server, HTML: Basic Tags, Text Formatting, Emphasizing Material in a Web Page, Text Styles, Spacing, Lists, Adding Graphics, Tables, Frames, Linking Documents.
2. Cascading Style Sheets: Introduction, Levels of style sheets, Style specification format, Selector, Forms, Property value form, Font properties, List properties, Colour, Alignment, Box model, Background Images, & tags.
3. JavaScript: Introduction, Object orientation and JavaScript, Syntactic features, Primitives, Operations and Expressions, Output & Input, Control statements, Object creation and modification, Arrays, Functions, Constructors, Pattern matching using regular expressions, JavaScript execution environment, Document Object Model, Element access in JavaScript, Event, Event handling- Body elements, Button elements, Textbox, Password elements. DOM2 event model.

UNIT II

4. XML: Introduction, Syntax, Document structure, Document type definition, Namespaces, XML schemas, Displaying raw XML documents, Displaying XML documents with CSS, XSLT style sheets, XML processor.
5. AJAX - Introduction, AJAX Components, Handling Dynamic HTML with Ajax, CSS to Define Look and Feel, AJAX using XML and XML Http Request- Introduction, Accessing, Creating and Modifying XML Nodes, Loading XML Data into an HTML Page, Receiving XML Responses, Handling Response XML.

6. Angular JS: MVC architecture, Directives, Expression, Controllers, Filters, Tables, Modules, Forms, Includes, Views, Scopes, Services, Dependency injection, Custom directives, Internationalization.

UNIT III

7. Python: Python variables, Python basic Operators, Python blocks, Data Types, Declaring and using Numeric data types: int, float, complex, Using string data type and string operations, Defining list and list slicing, Use of Tuple data type.

8. Program Flow Control: Conditional blocks using if, else and elif, for loops, for loop using ranges, string, list and dictionaries, while loops, Loop manipulation using pass, continue, break and else, Programming using conditional and loops block.

UNIT IV

9. Python Functions, Modules and Packages: Organizing python codes using functions, Organizing python projects into modules, Importing own module as well as external modules, Understanding Packages, Powerful Lambda function in python,

10.Python File Operation: Reading config files in python Writing, log files in python, Understanding read functions, read(), readline() and readlines(), Understanding write functions, write() and writelines(). Manipulating file pointer using seek Programming using file operations, Class, Exception handling, SQLite, Database connectivity,

11. Django Framework: Introduction, Web Development, Sample application.

Suggested Reading:

1. Deitel H.M., Deitel P.J.; Internet & World Wide Web: How to program; Pearson Education.
2. Kogent Learning; Web Technologies: HTML, JavaScript, PHP, Java, JSP, XML, AJAX – Black Book, Wiley India Pvt. Ltd.
3. Brad Dayley; Node.js, MongoDB, and AngularJS Web Development; Addison-Wesley Professional.
4. Sandeep Panda; Angular JS: Novice to Ninja; SPD, Indian Reprint.
5. Black book; Web Technologies: HTML, JS, PHP, Java, JSP, ASP.NET, XML and AJAX; Dreamtech Press.
6. Arun Ravindran, Samuel Dauzon, Aidas Bendoraitis; Django: Web Development with Python; Ingram short title.
7. Fabrizio Romano; Learn Web Development with Python: Get hands-on with Python Programming and Django web development; Packt Publishing Ltd.
8. William S Vincent; Django for Beginners: Build Websites with Python and Django; Welcometocode.
9. Martin C. Brown Python: The Complete Reference, TMH Publisher.

Paper Title: Analysis and Design of Algorithms

Paper Code: CS – 2119 **Max. Marks: 80 Time: 3 Hrs.**
Course Duration: 60 Lectures of one hour each.

Objective: The objective of the module is to create skills in students to design and analyze algorithms.

Note :
(i) The Question paper will consist of four units.
(ii) Examiner will set total of **nine** questions comprising two questions from each unit and one compulsory question of short answer type covering whole syllabi and having equal distribution of marks from all the units.
(iii) The students are required to attempt one question from each unit and the compulsory question.
(iv) All questions carry equal marks.

Course Outcomes :
After successful completion of course the students will be able to:

- Analysis of algorithms including recursive algorithms, space and time complexity.
- Understand the Divide and Conquer design strategy to solve different problems like sorting, selection and matrix multiplication.
- The students will also be able to various Dynamic Programming design techniques.
- Backtracking and Branch n Bound design techniques and application of these techniques to solve problems. Deal with finding lower bounds on complexity for sorting and search problems using comparison trees and the lower bounds on parallel computations. NP-Hard and NP-Complete problems and the examples of such problems able be written.

UNIT I

1. **Algorithms and Analysis :** Introduction, Algorithms specification, Recursive algorithms, space and time complexity, Asymptotic Notation (O,Ω, and Θ; o,ω), Practical complexities, Best, Average and worst case performance of algorithms, Examples, Introduction to recurrence relations: Solving recurrence relations using repeated substitutions.

2. **Divide and Conquer:** General method, Binary search, Merge sort, Quick sort, Selection problem, Matrix Multiplication using Divide and Conquer and Stassen's method and analysis of these problems.

UNIT II

3. **Greedy Method:** General Method, Knapsack problem, Job Sequencing with deadlines Problem, Minimum spanning Trees, Analysis of these problems.

4. **Dynamic Programming:** General method, Optimal binary search trees, 0/1 Knapsack, Travelling salesperson problem.

UNIT III

5. **Back Tracking :** General method, 8 queen's problem, Graph coloring, Hamiltonian cycles and Analysis of these problems.
6. **Branch-And-Bound :** General Method least cost search, upper bounding, 0/1 Knapsack and Travelling Salesperson problems, Dynamic state space organization for Travelling Salesperson problem.

UNIT IV

7. **Lower-Bound Theory:** Introduction to lower bounds, Comparison Trees, Lower bounds for ordered searching and sorting, Lower bounds on Parallel Computations: Lower bound on average case sorting.

8. **NP-hard and NP-complete problems:** Basic concepts, Deterministic and Non-Deterministic algorithms, NP-Hard and NP-Complete classes, Statement of Cook's Theorem, Examples of NP-hard graph and NP-Hard scheduling problems, NP-Hard Graph problem with proof: Clique Decision Problem.

Suggested Readings:

1. Horowitz, Ellis and Sahni, Sartaj: Fundamentals of Computer Algorithms, Universities Press Publications
2. Aho, A.V., Hopcroft, J.E., Ullman, J.D.: The Design and Analysis of Computer Algorithms, Addison-Wesley
3. Bentley, J.L.: Writing Efficient Programs, Prentice-Hall India, Eastern Economy Edition.
4. Goodman, S.E. &Hedetniemi, 2004: Introduction to the Design and Analysis of Algorithms, McGraw- Hill Book Comp.
5. Knuth, D. E., Fundamental of Algorithms: The Art of Computer Programming, Vol.-1, Naresh Publ. House.
6. Brassad, Gilles and Bartley, Paul 1996: Fundamentals of Algorithms, Prentice Hall of India.
7. Mark Allen Weiss: Data Structure and Algorithms Analysis in C++, Pearson Education.

PRACTICAL EXAMINATIONS (SESSION 2022-2023)
The Practical examination will be conducted for each of the following:

Title : OOPs using C++ and JAVA (Minor Project)
Paper Code : PR - 2115 (Based on CS - 2115)

Time : 3 Hrs.
Max. Marks : 100
External : 80
Internal : 20

Note : i) Each student is required to undergo 04 to 06 weeks Summer Training at the end of Second Semester.
The Internal Assessment Marks for each practical will be based on Minor Project.
ii) Paper will be set at the time of examination. Due weight-age may be given to the practical note-book and Assignments in evaluation.

Title : Web Technologies and Python Programming & Artificial Intelligence and Soft Computing (Minor Project)
Paper Code : PR - 2118 (Based on CS – 2118 & CS - 2124)

Time : 3 Hrs.
Max. Marks : 100
External : 80
Internal : 20

Note : i) Each student is required to undergo 04 to 06 weeks Summer Training at the end of Second Semester.
The Internal Assessment Marks for each practical will be based on Minor Project.
ii) Paper will be set at the time of examination. Due weight-age may be given to the practical note-book and Assignments in evaluation.

SEMESTER III

Paper Title: Interactive Computer Graphics

Paper Code: CS – 2120 **Max. Marks: 80 Time: 3 Hrs.**
Course Duration: 60 Lectures of one hour each.

Objective: The aim is to introduce the students to key concepts of Computer Graphics like display devices, co-ordinate system, transformations, line and circle drawing, pointing, positioning, projections, etc.

Note :
(i) The Question paper will consist of four units.
(ii) Examiner will set total of **nine** questions comprising two questions from each unit and one compulsory question of short answer type covering whole syllabi and having equal distribution of marks from all the units.
(iii) The students are required to attempt one question from each unit and the compulsory question.
(iv) All questions carry equal marks.

Course Outcomes :
After successful completion of course the students will be able to:

- After completion of course the student will be many basic understanding of the core concepts and the structure of an interactive computer graphics and its components and understanding of a typical graphics pipeline.
- Perform of geometrical transformations: 2D and 3D using OpenGL to create interactive computer graphics.
- A knowledge and understanding of interaction techniques.
- Create interactive graphics applications / minor projects using 'C' programming.

UNIT I

1. Display Devices :
Line and point plotting systems, Raster, vector, pixel and point plotters, Continual Refresh and storage displays, Digital frame buffer, Plasma panel displays, Display processors, Character generators, Colour-display techniques : shadow mask and penetration CRT, Colour look-up tables, Hard-copy colour printers.

UNIT II

2. Display Description:
Screen co-ordinates, User co-ordinates, Use of homogeneous coordinates, Display code generation, Graphical functions, the view algorithm, Two-dimensional transformation, Linedrawing, Circle drawing algorithms.

UNIT III

3. Interactive Graphics:
Pointing and positioning devices (cursor, light pen, digitizing tablet, the mouse, track balls), Interactive graphical techniques, Positioning, (Elastic or Rubber Band lines, Linking, zooming, panning, clipping, windowing, scissoring), Mouse Programming.

4. 3-D Graphics:
Wire-frame, perspective display, perspective depth, Projective transformations, Hidden line and surface elimination (Back-face removal algorithm).

UNIT IV

5. Turbo-C Graphic Language:
Primitives (constants, actions, operators, variables), Plotting and geometric transformations, Display subroutines, Concept of Animation, Saving, Loading and Printing graphics images from/to disk, Animated algorithms for Sorting, Towers of Hanoi.

6. Open GL:
Primitives of the language, interface with C/C++.

7. Programming Projects:
Two Dimensional Transformations, 3-dimensional Transformations, Interactive Graphical Techniques, GUI, Turbo C (Graphics Routines) is to be used as the standard teaching tool.

Suggested Readings:

1. Giloi, W.K., 1978: Interactive Computer Graphics, Prentice-Hall.
2. Newman, W., Sproul, R.F., 1980: Principles of Interactive Computer Graphics, McGraw-Hill.
3. Rogers, D.F., 1985: Procedural Elements for Computer Graphics, McGraw-Hill.
4. Harrington, S., 1983: Computer Graphics: A Programming Approach, Tata McGraw-Hill.
5. Foley, J.D., Van Dam A., 1982: Fundamentals of Interactive Computer Graphics, AddisonWesley.
6. Hearn, D., Baker, P.M., 1986: Computer Graphics, Prentice-Hall.
7. Tosijasu, L.K., 1983: Computer Graphics, Springer-Verlag.
8. Kelley Bootle: Mastering Turbo C. Galgotia.
9. Plastock, Roy, 1986: Theory & Problems of Computer Graphics, Schaum Series, Tata McGraw Hill.

Paper Title: Theory of Computations

Paper Code: CS – 2121 **Max. Marks: 80 Time: 3 Hrs.**
Course Duration: 60 Lectures of one hour each.

Objective: The goal of this course is to provide students with an understanding of basic concepts of Theory of Computation.

Note :
(i) The Question paper will consist of four units.
(ii) Examiner will set total of **nine** questions comprising two questions from each unit and one compulsory question of short answer type covering whole syllabi and having equal distribution of marks from all the units.
(iii) The students are required to attempt one question from each unit and the compulsory question.
(iv) All questions carry equal marks.

Course Outcomes :
After successful completion of course the students will be able to:

- Demonstrate advanced knowledge of formal computation and its relationship to languages
- Distinguish different computing languages and classify their respective types
- Recognize and comprehend formal reasoning about languages.
- Students will demonstrate knowledge of basic mathematical models of computation and describe how they relate to formal languages.

UNIT I

1. Recursive Languages:
Recursive definition, Alphabets, Language, Regular expression, Definitions of Finite state machine, Transition graphs, Deterministic & non-deterministic finite state machines, Regular grammar, Left-linear and right linear, Thomson's construction to convert regular.

2. Finite Automata : Expression to NDFA & subset algorithm to convert NDFA to DFA. Minimization of DFA, Finite state machine with output (Moore and Mealy Machine), Conversion of Moore machine to Mealy machine & vice-versa.

UNIT II

3. Properties of Regular languages:
Conversion of DFA to regular expression, Pumping lemma, Properties and limitations of finite state machine, Decision properties of regular languages, Applications of finite automata.

4. Context Free Grammar:
Context free grammar, Writing context free grammar for problems, Derivation tree and ambiguity, Application of context free grammars, Chomsky and Greibach Normal form, Conversion of CFG to CNF and GNF. Properties of context free grammar, CYK algorithm.

UNIT III

5. PDA:
Push down stack machine, Design of deterministic and non-deterministic push-down stack.

6. Turing Machine:
Turing machine definition and Design of Turing Machine, Church-Turing Thesis, Variations of Turing Machines, combining Turing machine, Universal Turing Machine, Post Machine, Chomsky Hierarchy.

UNIT IV

7. Compiler Structure : Precondition, post-condition, axiomatic approach for semantics, correctness, denotational semantics, compiler structure, compiler construction tools, compilation phases.

8. Parsing : YAKK package on UNIX system, Top Down Parser, Predictive Parser and Bottom up Parser.

Suggested Readings:

1. Lewis, Harry R. and Papadimitriou, Christos H.: Theory of Computation, Prentice Hall of India, | 1996.
2. Hopcroft, John E. and Ullman, Jefrey D.: Introduction to Automata Theory, Languages and | Computation, Addison-Wesley Publishing Company Inc
3. Brady, J.M.: Theory of Computer Science, Wiley.
4. Dewire, Dawna Tranis: Client Server Computing, McGraw Hill.
5. Aho,Lam,Sethi and Ullman : Compilers Principles, Techniques and Tools, Publisher Pearson.

Paper Title: Advance JAVA and Network Programming

Paper Code: CS – 2122 **Max. Marks: 80 Time: 3 Hrs.**
Course Duration: 60 Lectures of one hour each.

Objective: To create enterprise application development skills among students using Advanced Java.

Note :

(i) The Question paper will consist of four units.
(ii) Examiner will set total of **nine** questions comprising two questions from each unit and one compulsory question of short answer type covering whole syllabi and having equal distribution of marks from all the units.
(iii) The students are required to attempt one question from each unit and the compulsory question.
(iv) All questions carry equal marks.

Course Outcome
After successful completion of course the students will be able to:

- Development of interactive Desktop and Client/Server applications using UI widgets of Swing and RMI
- Access database through Java applications using Java Data Base Connectivity (JDBC)
- Development of dynamic web applications using Servlets and JSP
- Development of Network applications using TCP and UDP sockets

UNIT I

1. Swing : Features, components, Swing vs AWT, Swing containers, Controls, using Dialogs, Sliders, Progress bars, Tables, Event handling, Creating user interface using swing.

UNIT II

2. Java Database Connectivity : Connectivity model, Java.SQL package, JDBC Exception classes, Database connectivity, Data manipulation and navigation, creating database applications.
3. Java RMI : Distributed object technologies, RMI architecture, creating RMI applications.

UNIT III

4. Java Servlets : Servelets vs CGI, Servlet lifecycle, Creating and running simple servlets.
5. Networking : Networking basics, Client/server model, Java and the Net, TCP/IP client sockets, TCP/IP server sockets, Inet Address, URL, Data grams, Creating simple networking applications.

UNIT IV

6. Java Beans : Component architecture, Advantages of Beans, Bean Developer kit (BDK), JAR files, introspection, developing Beans, Using Bound properties, The Java Beans API, Introduction to EJB (Enterprise Java Beans),Types of EJB, Uses of EJB.

7. Java Server Pages : Introduction, JSP Architecture, JSP objects, Developing simple Web Applications.

Suggested Readings :

1. Schildt, Herbert : The Complete Reference Java 2, , TMH.
2. Ivan Bayross : Web Enabled Commercial Application Development using Java 2.0, BPB.
3. Cornell, Gary and Horstmann Cay S. : Core Java, Vol I and Vol II, Sun Microsystems Press.
4. Keogh, James : J2EE : The Complete Reference.
5. Martin Bond, Debbie Law, Andy Longshaw, Dan Haywood, Peter Roxburgh: Sams: Teach Yourself J2EE in 21 days, Pearson.

Paper Title: Mobile Communication and Application Development

Paper Code: CS – 2123 **Max. Marks: 80 Time: 3 Hrs.**
Course Duration: 60 Lectures of one hour each.

Objective: The course will familiarize the students with basic concepts about mobile communication, its architecture, protocols, mobile databases and operating systems. It will also enable them to develop mobile applications using Android.

Note :
(i) The Question paper will consist of four units.
(ii) Examiner will set total of **nine** questions comprising two questions from each unit and one compulsory question of short answer type covering whole syllabi and having equal distribution of marks from all the units.
(iii) The students are required to attempt one question from each unit and the compulsory question.
(iv) All questions carry equal marks.

Course Outcome :
After successful completion of course the students will be able to:

- Understanding of fundamental concepts of GSM, GPRS, CDMA and latest generations of Mobile Telephony
- Familiarization with Mobile database concepts such as hoarding, caching, data delivery, selecting tuning, indexing and various Mobile Operating systems
- Development of interactive Mobile applications using Android platform
- Testing and deployment of Mobile applications on virtual and physical devices

UNIT I

1. Mobile communication: Definition, Guided Transmission, Unguided Transmission; Mobile computing architecture, Mobile Devices, Mobile System Networks: Cellular, WLAN, Ad-hoc networks; GSM: Services and System Architecture, Radio Interfaces, Localization, Calling, Handover, Security; CDMA: Architecture, Spread Spectrum, Coding Methods ; GPRS: Introduction ,system Architecture; Introduction to EDGE.

UNIT II

2. Mobile database: Database hoarding techniques, Data Caching, Client Server computing: 2 tier and 3 tier client server architecture; Transactional models, Query processing, Data Recovery process; Data Dissemination; Communication Asymmetry, Classification of Data delivery mechanism: Push based, pull based, Hybrid; Selective tuning and indexing techniques, Mobile Application Languages, Mobile Operating system: Symbian, Android, iOS, Windows.

UNIT III

3. Android Application Development: Android Architecture, Getting started with Android, Mastering Android Development tools: Using Android Documentation, Working with Android Emulator; Building simple Android Applications: Using the Application Context, Working with

Activities, Using callback methods, Working with intents, Dialogs, Fragments, Logging application information.

UNIT IV

4. Android Application Development: Managing Application Resources: Working with Simple Resource values, Draw-able Resources, Layouts, Files; Configuring the Android Manifest file and basic application Settings, registering activities, Designating the launch activity, Managing Application permissions, Designing an application framework.

Suggested Readings:

1. Kamal, Raj,Mobile Computing ,Oxford Higher Education.
2. Jochen Schiller, "Mobile Communications", PHI/Pearson Education, Second Edition.
3. Uwe Hansmann, LotharMerk, Martin S. Nicklons and Thomas Stober, "Principles of Mobile Computing", Springer, New York, 2003.
4. RetoMeier,Professional Android 2 Application Development ,Wrox Publications.
5. HazysztofWesolowshi, "Mobile Communication Systems", John Wiley and Sons Ltd.
6. Shane Conder, Lauren Darcey: Sams Teach Yourself Android™ Application Development in 24 Hours.

Paper Title: Software Engineering and Project Management

Paper Code: CS - 2117 **Max. Marks : 80 Time : 3 Hrs.**
Course Duration: 60 Lectures of one hour each.

Objectives: The course aims to enable students to apply concepts of Software Engineering and help them learn its principles and methods including emerging practices and support tools. It also provide knowledge of Software Project Management and Quality Assurance.

Note :
(i) The Question paper will consist of four units.
(ii) Examiner will set total of **nine** questions comprising two questions from each unit and one compulsory question of short answer type covering whole syllabi and having equal distribution of marks from all the units.
(iii) The students are required to attempt one question from each unit and the compulsory question.
(iv) All questions carry equal marks.

Course Outcome
After successful completion of course the students will be able to:

- After completion of course the student will have fundamental knowledge of software engineering and will be able to work in diverse application domains.
- Demonstrate an understanding of and apply current theories, models, and techniques that provide a basis for the software lifecycle.
- The student will have ability to use the techniques and tools necessary for engineering practice in software development.
- Apply their foundations in software engineering to adapt to readily changing environments using the appropriate theory, principles and processes.

UNIT I

1. **Introduction:** Software crisis, Software Engineering Goals, SDLC, Software Process Models: Waterfall, Prototyping, Spiral, Software Requirements Specifications (SRS): Components and Structure. Software Project Management: Introduction, Decomposition techniques: Problem Based Estimation (LOC, FP); Empirical Estimation Models: COCOMO model; Risk in estimation.

UNIT II

2. **Project Scheduling:** Importance, Sequencing and scheduling activity; Project network diagrams: Network planning models, Critical path analysis, Software Design: Objectives, Structured Analysis and Design Tools: DFD, DD, Decision table, Quality Assurance, Software Quality Attributes, Factors Affecting Software Quality, Building Software Quality Assurance Plan and it's Components. Quality Management Principles, ISO 9000 Quality Standard, SEI Capability Maturity Model.

UNIT III

3. **Software Testing and Techniques:** Software Testing, Objectives, Static and Dynamic Analysis, Black-Box Testing and its Technique: Equivalence Class Partitioning, Boundary Value Analysis, Cause-Effect Graph, White-Box Testing and its Techniques: Basis Path Testing, Structural Testing, Logic Based Testing, Fault Based Testing, Software Testing Strategies, Debugging, Functional Testing, Systems and Acceptance Testing, Testing Metrics.

UNIT IV

4. **Software Maintenance:** Components of Software Maintenance Process, Types of software maintenance, Software maintenance process models, maintenance metrics, Reverse Engineering, System Configuration Management (SCM): Basic requirements for SCM System, SCM principles, planning and organizing for SCM, Benefits of SCM, Change Management, Version and Release Management.

Suggested Readings:

1. Pressman: Software Engineering, Tata McGraw Hill.
2. Sommerville, I., : Software Engineering, Narosa Publ. House.
3. Mall,Rajib, : Fundamentals of Software Engineering.
4. Jalote, Pankaj, : An Integrated Approach to Software Engineering, Narosa Publ.
5. Fairley, R.E., : Software Engineering Concepts, McGraw Hill.
6. James A O'Brien, George M Maracas, Ramesh Behl: Management Information Systems, McGraw Hill.
7. Walker Royce: Pearson Education, : Software Project Management.
8. Joel Henry, Pearson Education: Software Project Management.

PRACTICAL EXAMINATIONS (SESSION 2022-2023)
The Practical examination will be conducted for each of the following:

Title : Interactive Computer Graphics and Mobile Communication and Application Development (Minor Project)
Paper Code : PR – 2120 and 2123 (Based on CS – 2120 and CS - 2123)

Time : 3 Hrs.
Max. Marks : 100
External : 80
Internal : 20

Note : i) Paper will be set at the time of examination. Due weight-age may be given to the practical note-book and Assignments in evaluation.

Title : Advance JAVA and Network Programming (Minor Project)
Paper Code : PR - 2122 (Based on CS - 2122)

Time : 3 Hrs.
Max. Marks : 100
External : 80
Internal : 20

Note : i) Paper will be set at the time of examination. Due weight-age may be given to the practical note-book and Assignments in evaluation.

SEMESTER IV

GUIDELINES FOR SUBMISSION OF PROJECT REPORT (CS - 2125)

The report should consist of the following :

- Cover page including Project title, Name of the student, Name of the Department and Names of the Project Guides (both External and Internal).
- Acknowledgements.
- Certificates from company and department duly signed by external guide, chairman and internal guide.
- Contents with page numbers.
- Introduction (includes background and application or importance of the project).
- Objectives.
- System Analysis.

System Feasibility Study :

- Software requirement specifications.
- Design with system flowcharts and input/output design.
- Implementation and Testing
  - Hardware and software used.
  - Listing of well commented programs with result/output or detailed algorithms with input and output.

Further Scope of the Project :

- Bibliography.
- Appendices (any other information related to project).

Each student should observe the following norms while submitting the synopsis/thesis for the Project:-

(a) Use both sides of the paper instead of only single side.
(b) Use one and half interline spacing in the text (instead of double space).
(c) Stop using a blank sheet before the page, carrying figure or table.
(d) Try to insert figure/table in the text page itself (instead of using a fresh page for it, each time).

Students must consult/inform the internal guides regarding the progress of their work at least once in 20 days. It is the duty of the student to be in touch with his/her internal guide. The student must prepare 5 copies of the report including one copy for self. The remaining four are to be submitted before 31st May every year as per the following :

1. Main Library, 2. Department Library, 3. Internal Guide, 4. Company

One softcopy of the work is to be submitted to the concerned head of the dept./institution along with the report. The student must present his/her work in 15 minutes mainly focusing on his/her contribution with the help of slides followed by demonstration of the practical work done. The Project Viva will be completed before 15th June every year. Exact dates will be informed before 31st May every year. Project Viva will be conducted by an external examiner, internal examiner and the internal guide